

Cloud LearnX

Mathematics

Circle Theorems — Practice Worksheet

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Date: 01 August 2026

Topic: Circle Theorems

Instructions:

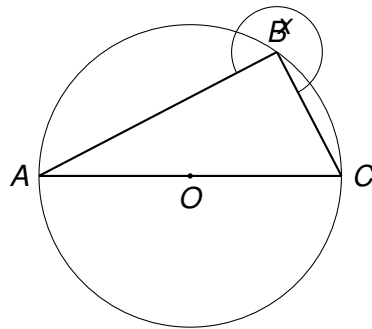
- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

Answer ALL ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

- 1** A space agency uses a circular satellite dish, shown below. Points A , B and C lie on the circumference of the circle with centre O , and AC is a diameter of the dish.



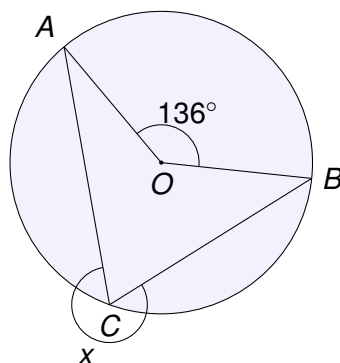
Find the size of angle x , giving a reason for your answer.

$x = \dots\dots\dots^\circ$

(2)

(Total for Question 1 is 2 marks)

- 2 Space engineers use circular cross-sections to model satellite dish reflectors. In the diagram below, O is the centre of the circle, and A , B , C are points on the circumference. The angle at the centre, $\angle AOB$, is 136° .



Work out the size of angle x , marked at point C on the circumference.

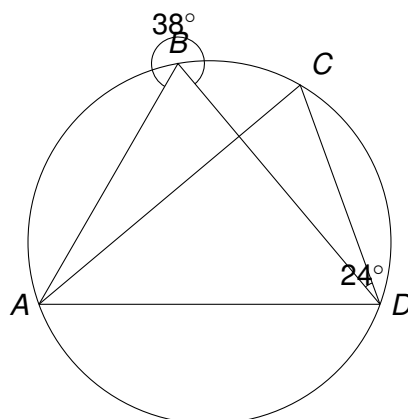
$$x = \dots\dots\dots^\circ$$

(2)

(Total for Question 2 is 2 marks)

- 3** Points A , B , C and D lie on the circumference of a circle, as shown in the diagram. Chord AD is drawn, and points B and C lie on the same side of AD .

Angle $ABD = 38^\circ$ and angle $BDC = 24^\circ$.



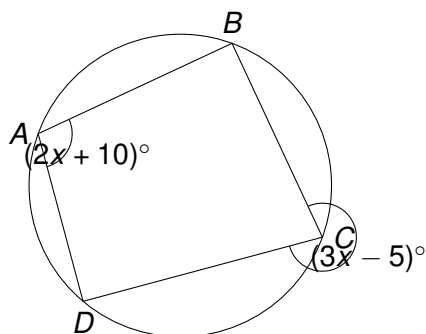
Given that angle $ABD = 38^\circ$ and angle $BDC = 24^\circ$, find the size of angle ACD , giving reasons for your answer.

$ACD = \dots\dots\dots^\circ$

(3)

(Total for Question 3 is 3 marks)

- 4 An archaeologist is studying the circular floor plan of an ancient observatory once used to track the stars. Four support pillars, at points A , B , C and D , lie on the circular boundary wall, forming the cyclic quadrilateral $ABCD$ shown below.



Angle $DAB = (2x + 10)^\circ$ and angle $DCB = (3x - 5)^\circ$.

- (a) Form an equation in x and solve it to find the value of x .

$x = \dots\dots\dots$

(2)

- (b) Work out the size of angle DAB and the size of angle DCB , giving a reason for your method.

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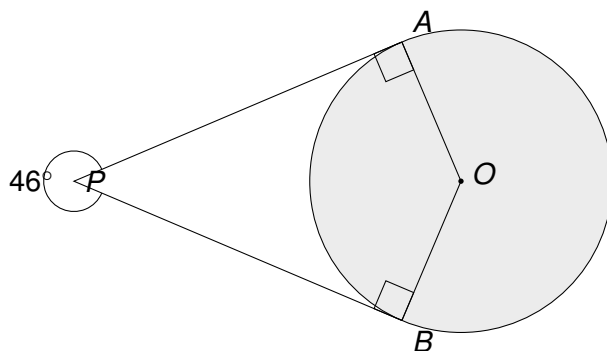
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(2)

(Total for Question 4 is 4 marks)

- 5 A space engineer is modelling the signal range of a satellite dish. The circular boundary of the signal, centre O , has two straight-line signal paths PA and PB drawn from the receiver at point P (outside the circle) to points A and B on the edge of the boundary. Each signal path just touches the circle, so PA and PB are tangents to the circle at A and B respectively.

Angle $APB = 46^\circ$.



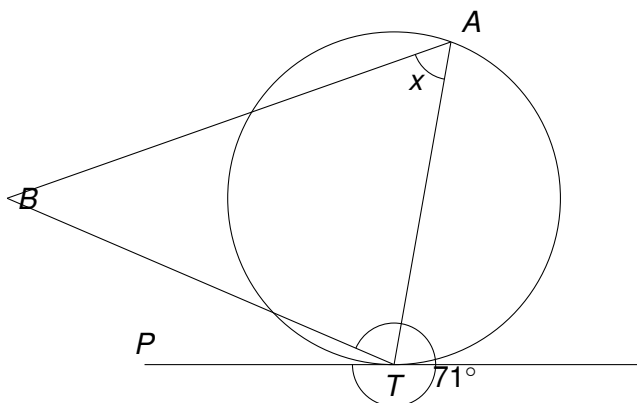
Find the size of angle AOB , the angle at the centre of the circle between the two radii OA and OB .

$AOB = \dots\dots\dots^\circ$

(4)

(Total for Question 5 is 4 marks)

- 6 A museum is designing a circular observation window for a planetarium exhibit on planetary orbits. The diagram shows the circular window, with PT a tangent to the circle at point T . TA and TB are chords of the circle, and A , B , T lie on the circumference. The angle between the tangent PT and the chord TB is 71° .



Show that the angle TAB , marked x on the diagram, is equal to 71° .

You must give full reasons for your answer.

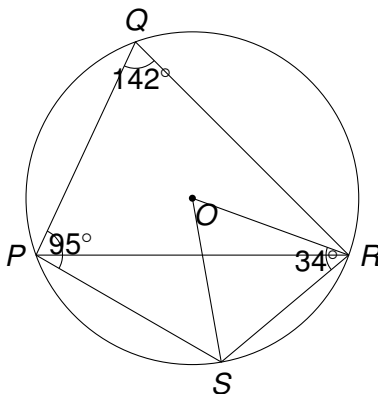
(4)

(Total for Question 6 is 4 marks)

- 7 A space agency tracks four satellites, P , Q , R and S , which orbit at a fixed distance from a control station O , so all four satellites lie on a circle with centre O . Points P , Q , R and S lie on the circumference, in that order around the circle.

The signal path PR passes through the region near O , and radio links are shown between P , Q , R and S as chords of the circle.

Angle $PQR = 142^\circ$ and angle $QPS = 95^\circ$. Angle $SRO = 34^\circ$, where O is the centre of the circle.



- (a) Work out the size of angle POR (the reflex angle at the centre corresponding to arc PQR), giving a reason for your answer.

$$POR = \dots\dots\dots^\circ$$

(2)

- (b) Work out the size of angle PSR , giving a reason.

$$PSR = \dots\dots\dots^\circ$$

(2)

- (c) Hence work out the size of angle ORS ...

(Use your answer to part (b) together with angle $SRO = 34^\circ$ to find angle PRS .)

$$PRS = \dots\dots\dots^\circ$$

(2)

(Total for Question 7 is 6 marks)

TOTAL FOR PAPER IS ?? MARKS